CIS 215 LAB REPORT

Network Security Fundamentals

6.3.6 Lab — Basic Device Configuration and OSPF Authentication

BACKGROUND	
Course:	CIS 215 - Network Security Fundamentals
Term:	Spring 2026
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Week / Module Reference:	Week 6 / OSPF Authentication
Assignment Reference:	Lab 6.3.6 / Packet Tracer Activity
Lab Type:	Hands-On Lab / Packet Tracer
Date Completed:	2026-04-20
Time Spent:	Approximately 2.0 hours

ENVIRONMENT

Platforms, Devices, and Systems Used:

- Primary host system: Windows desktop environment with Cisco Packet Tracer used for the lab build, testing, and verification.
- Lab platform: Cisco Packet Tracer simulation with the topology created manually from a blank workspace.
- Devices used in the topology: routers R1, R2, and R3; switches S1 and S3; hosts PC-A and PC-C.
- Router platform used in the simulation: ISR4321; switch platform used: 2960-24TT.
- Security focus area: control-plane hardening through OSPF authentication using SHA256 key chains.

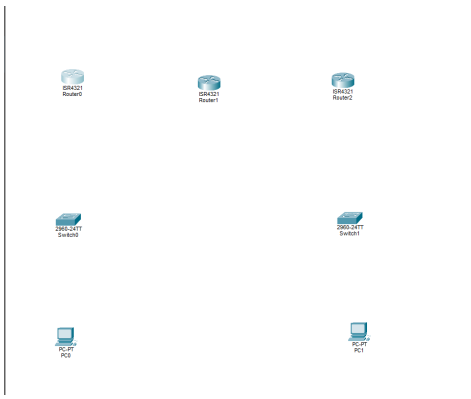

Figure 1. Initial Packet Tracer workspace after the required devices were placed on the board.

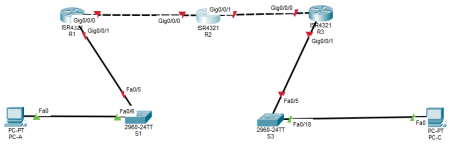

Figure 2. Topology after devices were renamed and wired according to the lab port mapping.

Files, Media, and Resources Used:

- Packet Tracer source file: completed 6.3.6 Basic Device Configuration and OSPF Authentication .pkt lab file.
- Lab instructions: assignment sheet containing the topology, addressing table, routing tasks, and authentication tasks.
- Evidence collected: topology screenshots, routing verification screenshots, troubleshooting screenshots, authentication verification screenshots, ping verification screenshots, and final topology screenshot.
- Supporting notes: addressing recap, per-router command tracking, CLI validation results, and troubleshooting notes.

Work Objective:

- Build the required three-router network from scratch and cable every device according to the provided topology.
- Configure router interfaces and host addressing, enable OSPF routing, and verify end-to-end communication across the network.
- Apply OSPF SHA256 authentication to secure the routing control plane and verify that routing remains stable after hardening.

Learning Objectives:

- Configure interface addressing on multiple routers from a supplied addressing table.
- Deploy OSPF process 1 and verify neighbors and learned routes across a transit topology.
- Use passive interfaces correctly on LAN-facing links while still advertising those networks into OSPF.
- Create an OSPF key chain and apply SHA256 authentication to OSPF-facing interfaces.
- Validate routing and security changes with show commands and end-to-end connectivity tests.

Scope and Constraints:

- In scope: topology build, router addressing, PC addressing, OSPF configuration, passive interfaces, connectivity tests, configuration saves, key chain creation, SHA256 authentication, and final verification.
- The work was completed entirely inside Cisco Packet Tracer using manual device placement, manual cabling, and router CLI work.
- The lab required a clean, working network before security changes could be introduced, so routing validation was completed first and authentication was added only after the baseline path was confirmed.

Addressing Plan Used:

- R1 G0/0/0: 10.1.1.1 / 255.255.255.252
- R1 G0/0/1: 192.168.1.1 / 255.255.255.0
- R2 G0/0/0: 10.1.1.2 / 255.255.255.252
- R2 G0/0/1: 10.2.2.2 / 255.255.255.252
- R3 G0/0/0: 10.2.2.1 / 255.255.255.252
- R3 G0/0/1: 192.168.3.1 / 255.255.255.0
- PC-A NIC: 192.168.1.3 / 255.255.255.0 / default gateway 192.168.1.1
- PC-C NIC: 192.168.3.3 / 255.255.255.0 / default gateway 192.168.3.1

TECHNICAL WORK PERFORMED

Preparation Performed:

- Reviewed the topology and addressing plan before beginning the build so that interface naming, cable placement, and host configuration would match the target design from the start.
- Placed the routers, switches, and PCs on the Packet Tracer board, then renamed each device to match the logical roles used throughout the lab instructions and CLI workflow.
- Mapped the cable plan before clicking connections in Packet Tracer so the final build would align with the addressing table and reduce rework later during routing verification.

Topology and Basic Configuration Procedure:

1) Placed and renamed R1, R2, R3, S1, S3, PC-A, and PC-C in the Packet Tracer workspace.
2) Cabled R1-R2, R2-R3, R1-S1, PC-A-S1, R3-S3, and PC-C-S3 using the assigned port mapping.
3) Configured hostnames on all routers, disabled DNS lookup, and applied interface addressing from the table.
4) Issued no shutdown on all active router interfaces and verified interface state with show ip interface brief.
5) Configured PC-A and PC-C with static IP addressing and the correct default gateways.
6) Rechecked the topology after the interfaces turned green and confirmed all physical links were active.

Routing Configuration Procedure:

1) Configured router ospf 1 on all three routers.
2) Added the 10.1.1.0/30 and 192.168.1.0/24 networks to OSPF on R1.
3) Added the 10.1.1.0/30 and 10.2.2.0/30 networks to OSPF on R2.
4) Added the 10.2.2.0/30 and 192.168.3.0/24 networks to OSPF on R3.
5) Marked G0/0/1 as passive on R1 and R3 so the LANs were advertised without attempting OSPF adjacency with hosts.
6) Validated neighbors with show ip ospf neighbor and validated learned routes with show ip route.

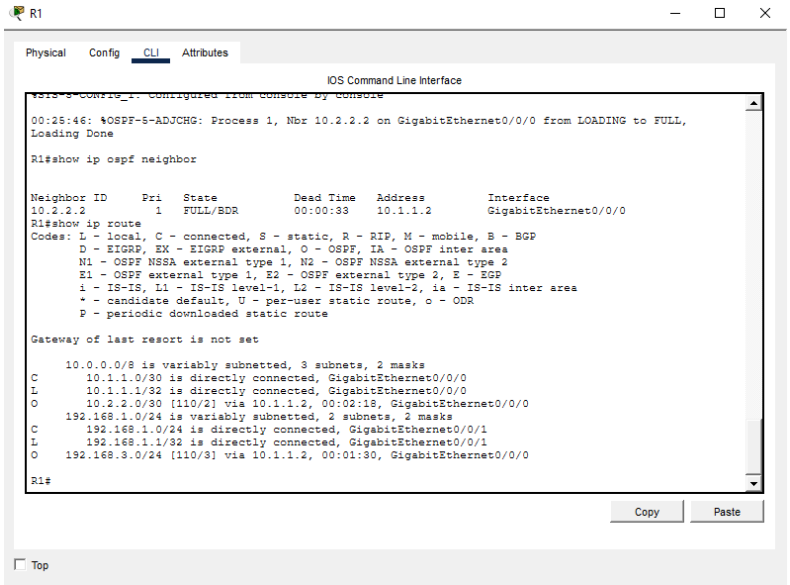

Figure 3. R1 OSPF verification showing a FULL neighbor adjacency and learned OSPF routes.

Connectivity Validation Procedure:

1) Verified that OSPF had formed correctly by confirming a FULL neighbor relationship between adjacent routers.
2) Confirmed that OSPF routes appeared in the routing table as remote LAN and transit entries.
3) Tested router-to-router reachability across the routed path before moving to host validation.
4) Tested PC-A to PC-C end-to-end connectivity and repeated the ping after ARP resolution settled.
5) Saved the working routing configuration on R1, R2, and R3 before moving into authentication.

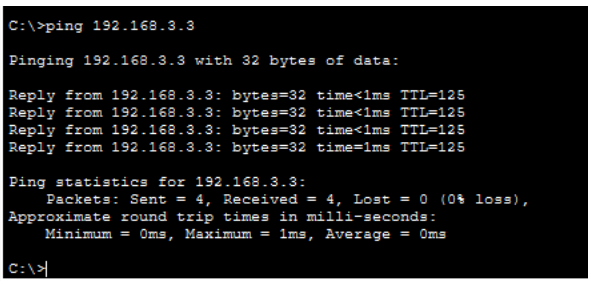

Figure 4. Successful end-to-end ping from PC-A to PC-C after routing and ARP resolution stabilized.

OSPF SHA256 Authentication Procedure:

1) Created key chain NetAcad on all three routers.
2) Created key 1 and assigned the key string NetSeckeystring.
3) Applied the cryptographic algorithm hmac-sha-256 under the key on all routers.
4) Applied ip ospf authentication key-chain NetAcad to R1 G0/0/0, R2 G0/0/0, R2 G0/0/1, and R3 G0/0/0.
5) Observed the expected temporary OSPF neighbor drop while authentication was only partially configured.
6) Re-verified show ip ospf interface, show ip ospf neighbor, and show ip route after all routers matched.

Final Validation and Verification:

- Verified that the OSPF interfaces reported cryptographic authentication enabled and HMAC-SHA-256 in use.
- Confirmed neighbors returned to FULL state after authentication was configured on every required interface.
- Confirmed OSPF routes remained present in the routing table after control-plane hardening.
- Confirmed PC-A could still successfully ping PC-C after authentication was enabled.
- Saved the final Packet Tracer topology in a completed, submission-ready state.

KEY COMMANDS, TROUBLESHOOTING, AND EVIDENCE

Key Commands:

BASIC ROUTER SETUP

```
enable
configure terminal
hostname R1
hostname R2
hostname R3
no ip domain-lookup
interface g0/0/0
interface g0/0/1
ip address 10.1.1.1
255.255.255.252
ip address 10.1.1.2
255.255.255.252
ip address 10.2.2.1
255.255.255.252
ip address 10.2.2.2
255.255.255.252
ip address 192.168.1.1
255.255.255.0
ip address 192.168.3.1
255.255.255.0
no shutdown
show ip interface brief
```

OSPF / VERIFY

```
router ospf 1
network 192.168.1.0 0.0.0.255
area 0
network 10.1.1.0 0.0.0.3 area 0
network 10.2.2.0 0.0.0.3 area 0
network 192.168.3.0 0.0.0.255
area 0
passive-interface g0/0/1
show ip ospf neighbor
show ip route
show run | section ospf
ping 10.2.2.1
ping 192.168.1.1
ping 192.168.1.3
ping 192.168.3.3
ping 192.168.3.3 -n 10
copy running-config
startup-config
```

AUTHENTICATION

```
key chain NetAcad
key 1
key-string NetSeckeystring
cryptographic-algorithm
hmac-sha-256
ip ospf authentication key-chain
NetAcad
show ip ospf interface g0/0/0
show ip ospf interface g0/0/1
show ip ospf neighbor
show ip route
end
exit
```

Problems Encountered and Fixes Applied:

- The first OSPF validation attempt showed no neighbors and no OSPF routes because the full OSPF configuration had not yet been applied consistently across all three routers.
- The issue was corrected by reapplying the exact OSPF network statements on R1, R2, and R3 and then rechecking neighbors.
- The first PC-A to PC-C ping failed even though routing was already working because ARP resolution had not fully settled.
- The issue was confirmed by successful local gateway and router-to-host pings, then resolved by repeating the end-to-end ping.

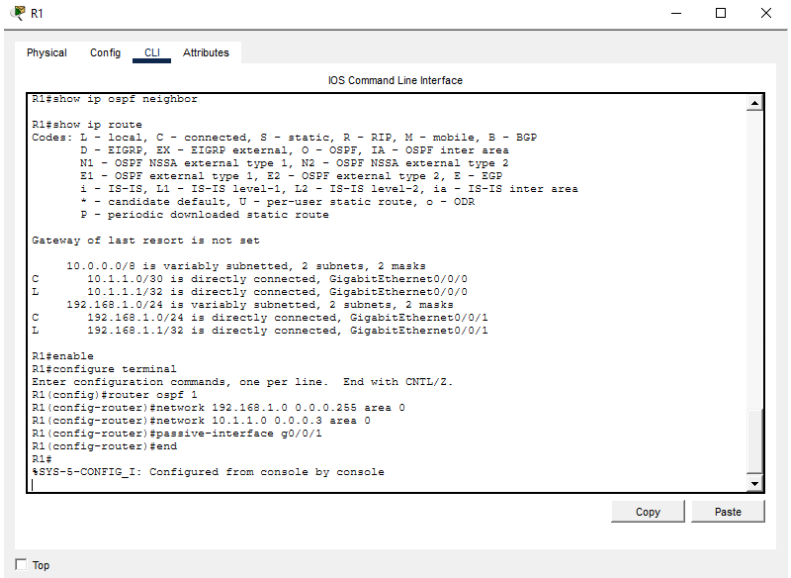

Figure 5. Early R1 verification showing that OSPF neighbors and learned routes were not yet present.

Evidence Collected During the Lab:

1) Topology screenshots documented device placement, renaming, cabling, and the final clean green-link topology.
2) Router CLI screenshots documented interface state changes, OSPF configuration, and routing-table validation.
3) Troubleshooting screenshots documented the stage where OSPF neighbors were absent and the later corrected FULL state.
4) Ping-test screenshots documented host-to-host success after routing and ARP resolution were complete.
5) Authentication screenshots documented SHA256 key-chain application, authenticated interfaces, FULL neighbors, and preserved routing after hardening.

Operational Notes / Observations:

- Link-state routing work is highly sequence-dependent; basic addressing and interface activation must succeed first.
- OSPF neighbors only formed when the network statements and area assignments matched correctly on all participating routers.
- Applying OSPF authentication one side at a time causes a temporary adjacency drop, which is expected behavior.
- Packet Tracer end-host connectivity may require an extra ping cycle after routing comes up because ARP must populate first.

Artifacts and Deliverables:

- Completed Packet Tracer .pkt lab file containing the fully working and authenticated topology.
- Supporting screenshots retained as evidence of the topology build, OSPF routing, troubleshooting, and final verification.
- Final written report documenting the completed lab workflow, technical outcomes, and validation steps.

SUMMARY, RESULTS, AND REFLECTION

Completion Verification:

- The topology was built manually and all required router, switch, and host connections were completed correctly.
- Router addressing, PC addressing, OSPF routing, passive interfaces, and end-to-end host communication were verified.
- The OSPF control plane was secured using key chain NetAcad, key 1, key-string NetSeckeystring, and HMAC-SHA-256.
- Authenticated OSPF neighbors returned to FULL state and the routing table remained correct after hardening.
- The final Packet Tracer file was saved in a working, submission-ready state after all validation steps were complete.

```
00:56:54: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.1.1 on GigabitEthernet0/0/0 from EXSTART to DOWN, Neighbor Down: Interface down or detached

00:56:57: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.1.1 on GigabitEthernet0/0/0 from LOADING to FULL, Loading Done

R2#show ip ospf interface g0/0/0

GigabitEthernet0/0/0 is up, line protocol is up
  Internet address is 10.1.1.2/30, Area 0
  Process ID 1, Router ID 10.2.2.2, Network Type BROADCAST, Cost: 1
  Transmit Delay is 1 sec, State BDR, Priority 1
  Designated Router (ID) 192.168.1.1, Interface address 10.1.1.1
  Backup Designated Router (ID) 10.2.2.2, Interface address 10.1.1.2
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
    Hello due in 00:00:01
  Index 1/1, flood queue length 0
  Next 0x0(0)/0x0(0)
  Last flood scan length is 1, maximum is 1
  Last flood scan time is 0 msec, maximum is 0 msec
  Neighbor Count is 1, Adjacent neighbor count is 1
    Adjacent with neighbor 192.168.1.1  (Designated Router)
  Suppress hello for 0 neighbor(s)
  Cryptographic authentication enabled
    Sending SA: Key 1, Algorithm HMAC-SHA-256 - key chain NetAcad
R2#show ip ospf interface g0/0/1

GigabitEthernet0/0/1 is up, line protocol is up
  Internet address is 10.2.2.2/30, Area 0
  Process ID 1, Router ID 10.2.2.2, Network Type BROADCAST, Cost: 1
  Transmit Delay is 1 sec, State DR, Priority 1
  Designated Router (ID) 10.2.2.2, Interface address 10.2.2.2
  Backup Designated Router (ID) 192.168.3.1, Interface address 10.2.2.1
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
    Hello due in 00:00:01
  Index 1/2, flood queue length 0
  Next 0x0(0)/0x0(0)
  Last flood scan length is 1, maximum is 1
  Last flood scan time is 0 msec, maximum is 0 msec
  Neighbor Count is 1, Adjacent neighbor count is 1
    Adjacent with neighbor 192.168.3.1  (Backup Designated Router)
  Suppress hello for 0 neighbor(s)
  Cryptographic authentication enabled
    Sending SA: Key 1, Algorithm HMAC-SHA-256 - key chain NetAcad
R2#show ip ospf neighbor

Neighbor ID     Pri   State           Dead Time   Address         Interface
192.168.1.1       1   FULL/DR         00:00:33    10.1.1.1        GigabitEthernet0/0/0
192.168.3.1       1   FULL/BDR        00:00:31    10.2.2.1        GigabitEthernet0/0/1
R2#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 4 subnets, 2 masks
C       10.1.1.0/30 is directly connected, GigabitEthernet0/0/0
L       10.1.1.2/32 is directly connected, GigabitEthernet0/0/0
C       10.2.2.0/30 is directly connected, GigabitEthernet0/0/1
L       10.2.2.2/32 is directly connected, GigabitEthernet0/0/1
O    192.168.1.0/24 [110/2] via 10.1.1.1, 00:01:08, GigabitEthernet0/0/0
O    192.168.3.0/24 [110/2] via 10.2.2.1, 00:01:42, GigabitEthernet0/0/1

R2#
```

Figure 6. R2 authentication verification showing SHA256 enabled on both OSPF-facing interfaces and both neighbors FULL.

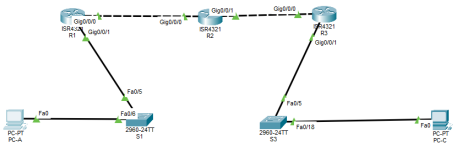

Figure 7. Final topology with the routed network fully operational and all required links in the completed state.

Key Findings:

1) Reliable dynamic routing depends on accurate interface addressing, correct OSPF network statements, and consistent area use.
2) Passive interfaces are important because they allow LAN advertisement without wasting neighbor attempts on host segments.
3) OSPF authentication protects the routing control plane and must match exactly on all participating interfaces.
4) Troubleshooting is faster when validation is done in layers: interfaces first, routing second, host reachability third.

Security Concepts Reinforced:

- Control-plane security matters because routing protocols can be a direct attack surface if left unauthenticated.
- Shared key chains and modern cryptographic algorithms such as HMAC-SHA-256 strengthen routing trust between routers.
- Authentication must match on every participating interface or adjacencies will fail, routes will disappear, and connectivity will break until the mismatch is corrected.
- Verification commands are part of security work because configuration without confirmation is not enough.

Best Practices / Hardening Notes:

- Complete and verify the unsecured routing baseline before introducing authentication so failures are easier to isolate.
- Validate every major stage immediately after it is configured rather than waiting until the very end of the lab.
- Use show ip ospf neighbor, show ip route, and show ip ospf interface together because each command validates a different part of the control-plane and data-plane picture.
- Save router configurations after major milestones so a working state is preserved before introducing additional changes.

Reflection:

- This lab reinforced that network security is not only about user access but also about protecting routing behavior between devices.
- The most useful takeaway was seeing a complete workflow from blank topology, to working OSPF, to authenticated OSPF, and then validating that the network still functioned end to end after security controls were applied.
- From a professional standpoint, the finished work demonstrates hands-on ability to build a routed topology, validate it, troubleshoot it, and harden the control plane without breaking operations.

Next Steps:

- Attach the completed Packet Tracer file and submit it through Canvas as required by the assignment.
- Store the final screenshots and report in the course repository as polished evidence of the completed lab.
- Reuse the same phased workflow in future labs that involve stronger routing controls, larger topologies, and deeper network verification.